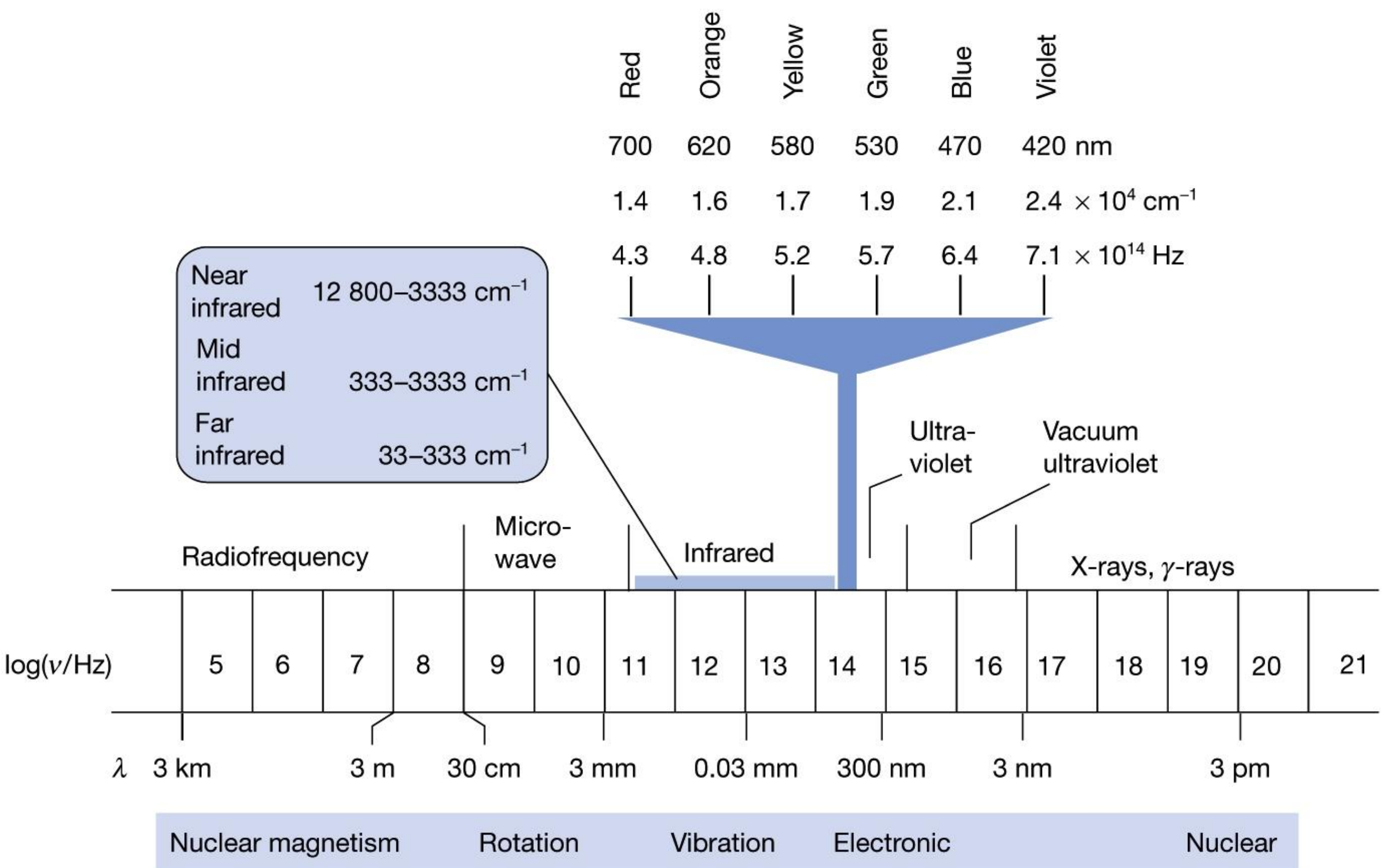
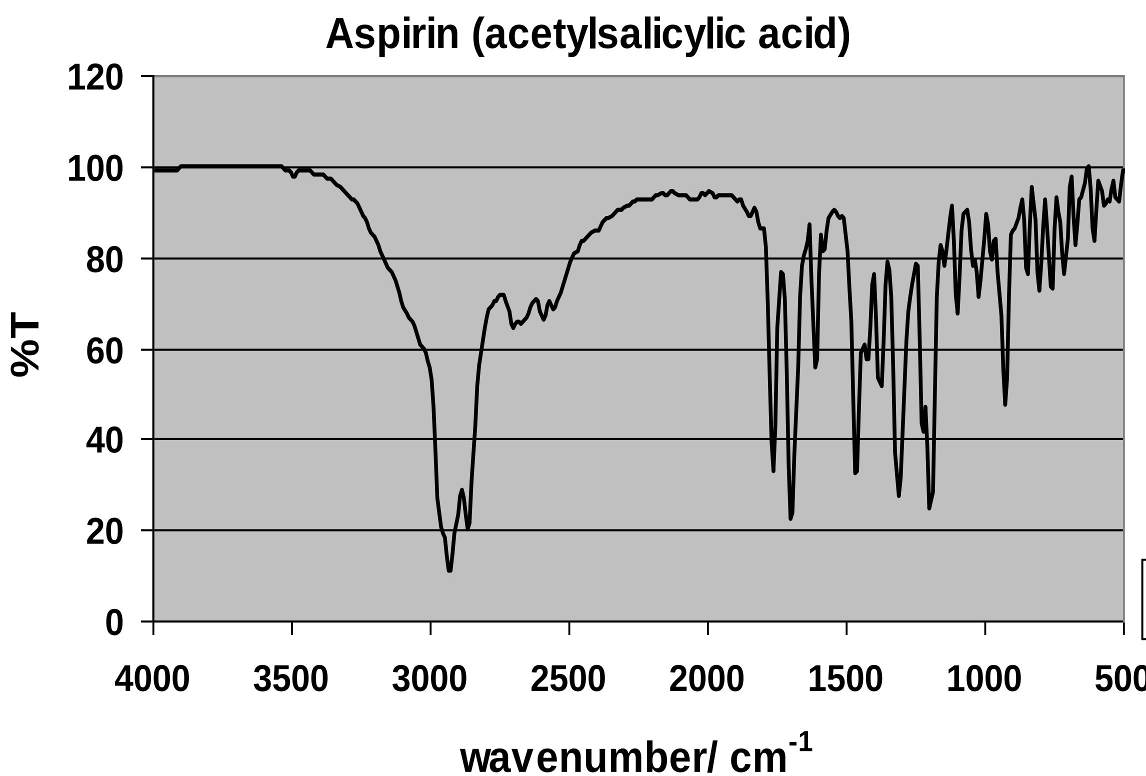
Lecture 02

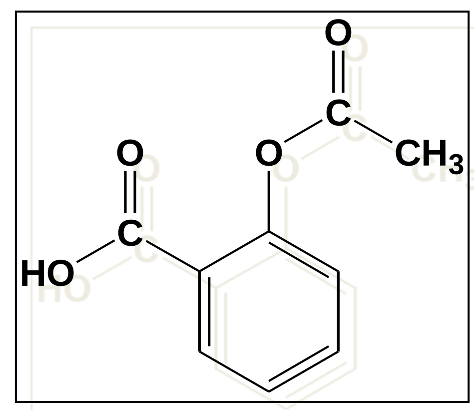
INFRARED SPECTROSCOPY

THE ELECTROMAGNETIC SPECTRUM



- Infrared radiation stimulates molecular vibrations.
- Infrared spectra are traditionally displayed as %T (percent transmittance) versus wave number (4000-400 cm-1).
- Useful in identifying presence or absence of functional groups.

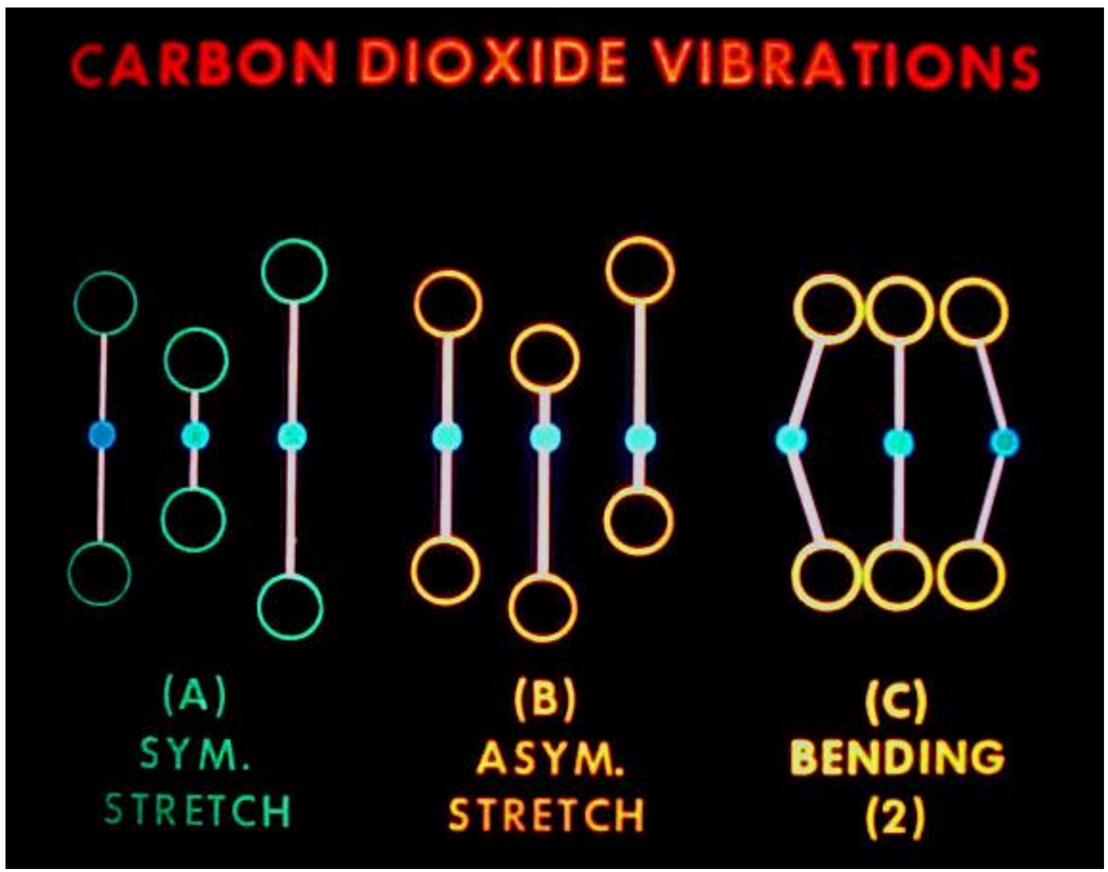




Wavenumber = 1/ wavelength

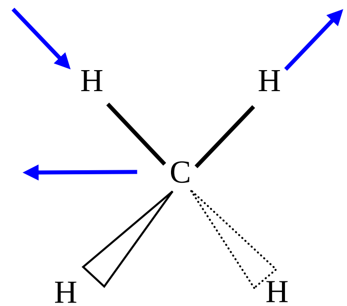
INFRARED SPECTROSCOPY

- In the IR region of the electromagnetic spectrum, the absorption of radiation by a sample is due to changes in the vibrational energy states of a molecule.

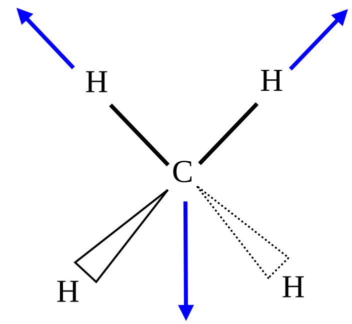


INFRARED SPECTROSCOPY

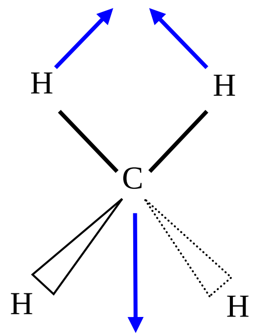
Methane



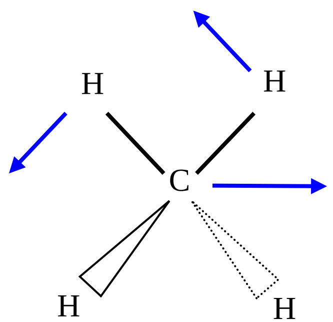
Asymmetrical stretching



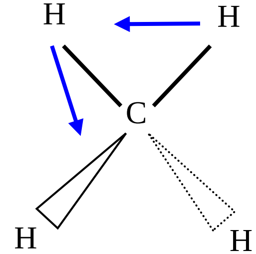
Symmetrical stretching



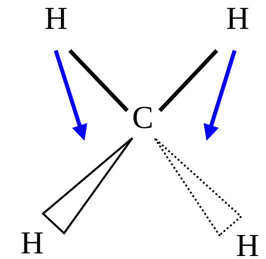
Bending or scissoring



Rocking or in plane bending



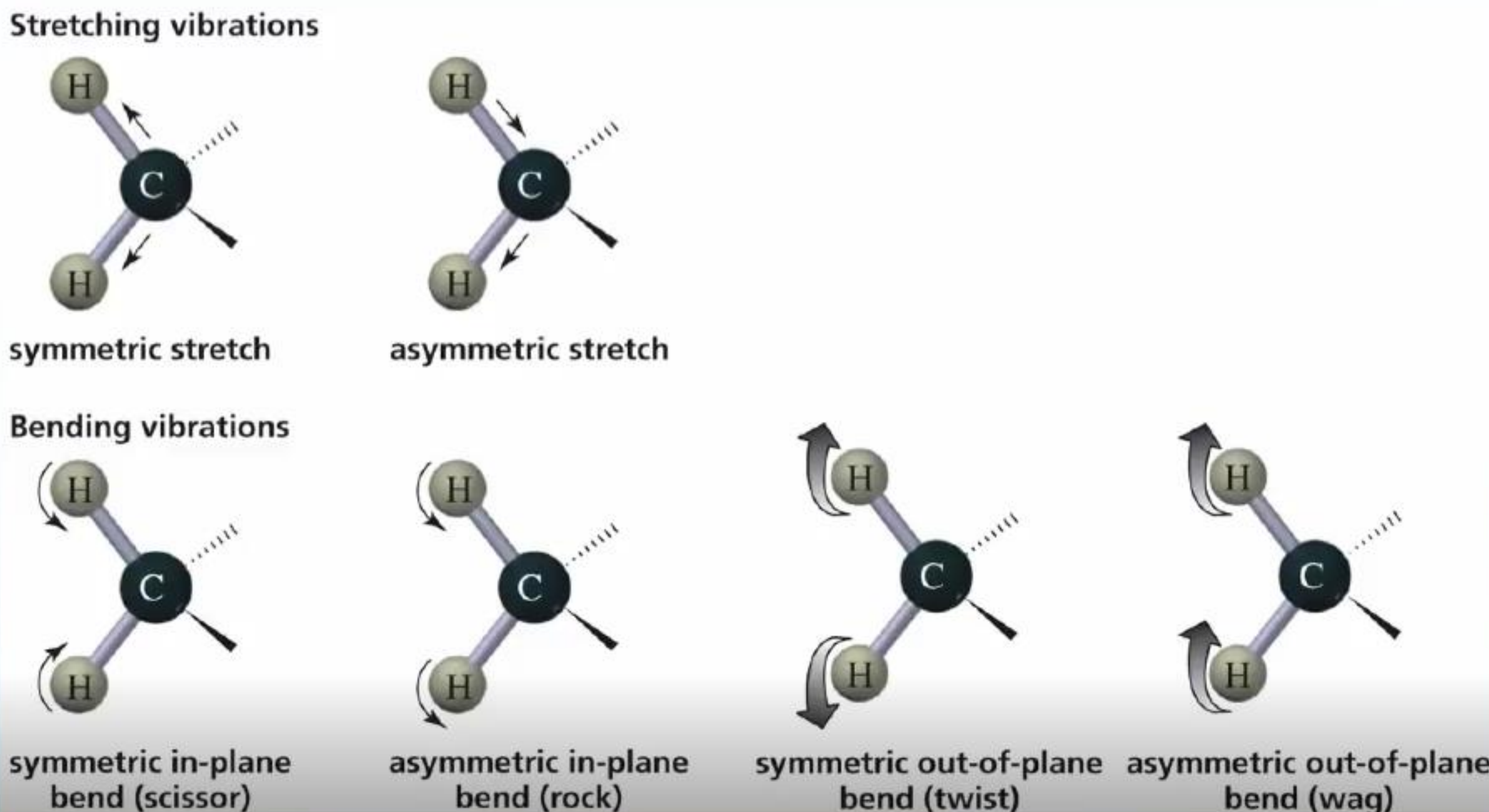
Twisting or out-of-plane bending



Wagging or out-of-plane bending

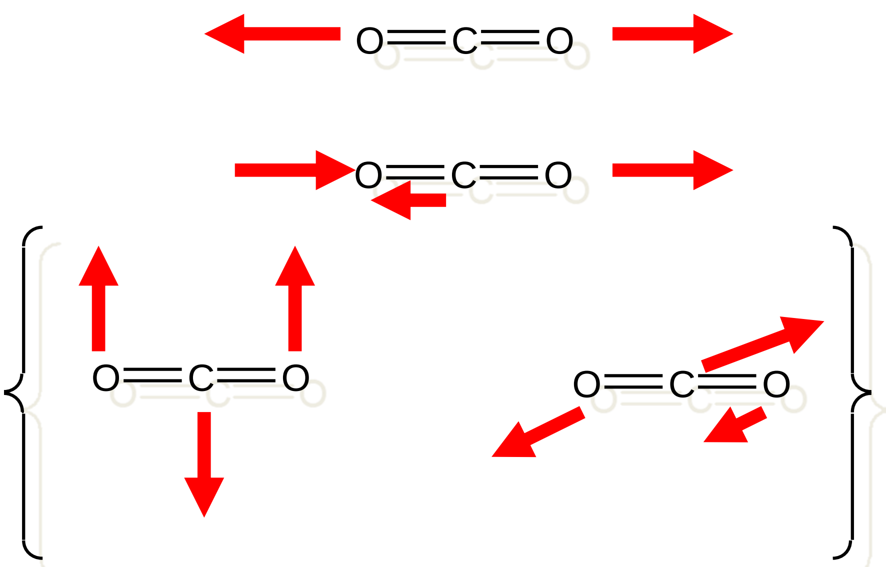
General Theory of IR Spectroscopy

Normal modes of vibration = stretching and bending



INFRARED SPECTROSCOPY

- Only vibrations that cause a change in 'polarity' give rise to bands in IR spectra



Symmetric stretch

Asymmetric stretch

Bending (doubly degenerate)

INFRARED SPECTROSCOPY

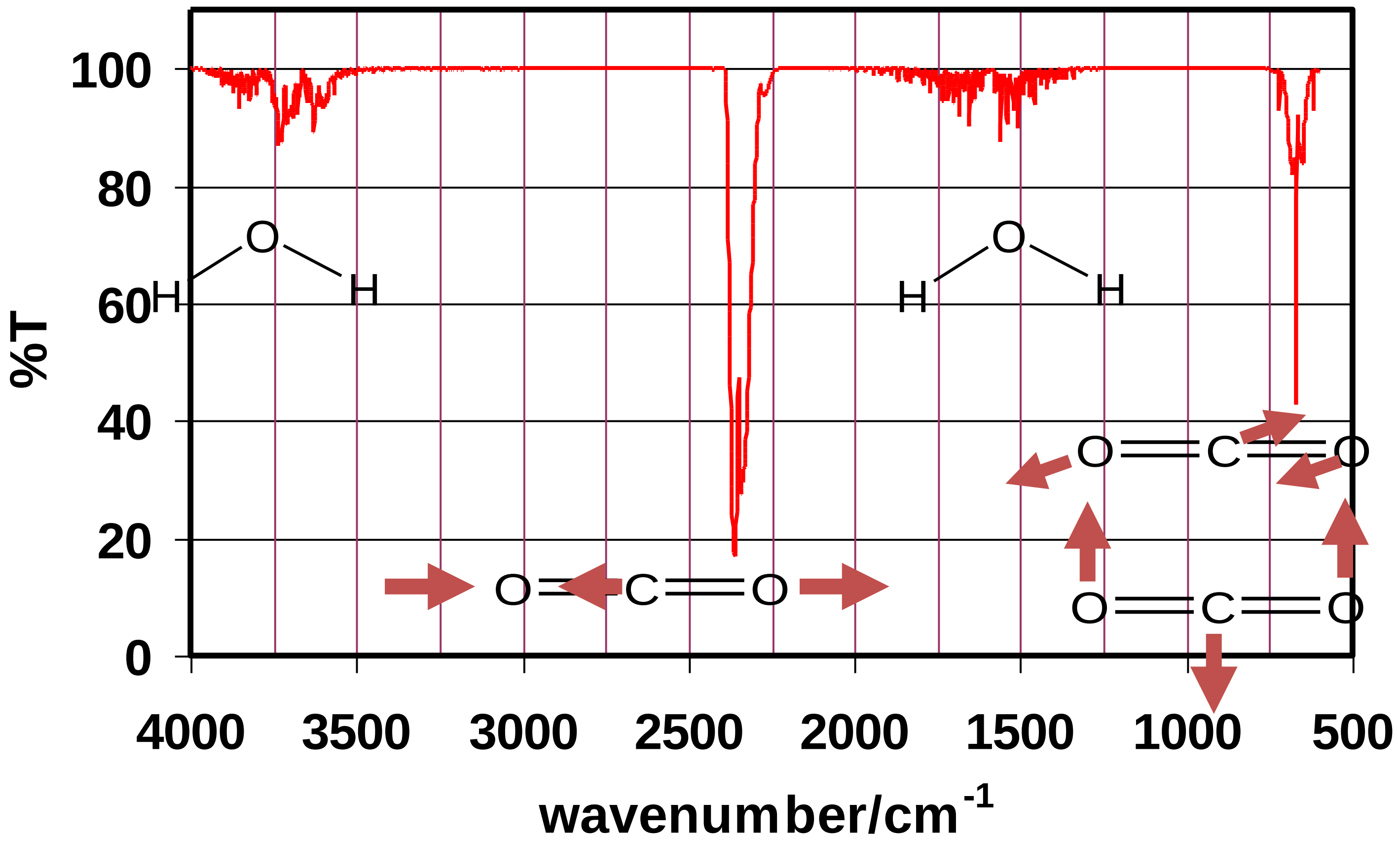
- Any change in shape of the molecule- stretching of bonds, bending of bonds, or internal rotation around single bonds

- Asymmetrical stretching/bending and internal rotation change the dipole moment of a molecule. Asymmetrical stretching/bending are IR active.

- Symmetrical stretching/bending does not. Not IR active

INFRARED SPECTROSCOPY

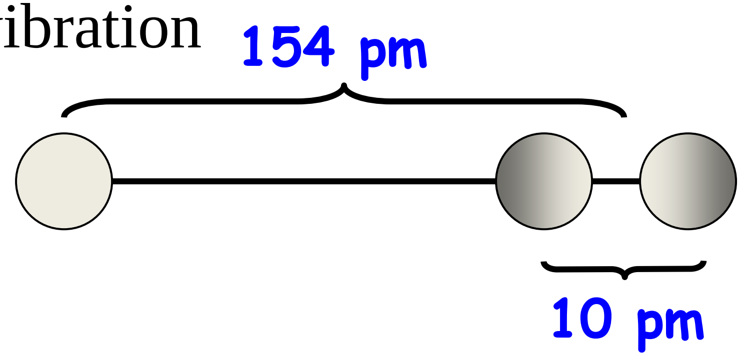
- Human Breath



INFRARED SPECTROSCOPY

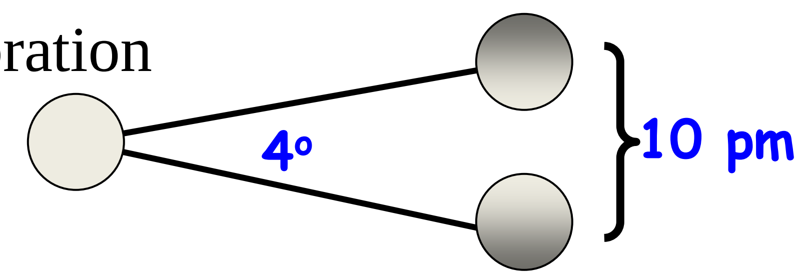
- How much movement occurs in the vibration of a C-C bond?

stretching vibration



For a C-C bond with a bond length of 154 pm, the variation is about 10 pm.

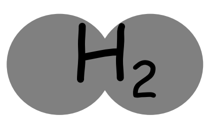
bending vibration

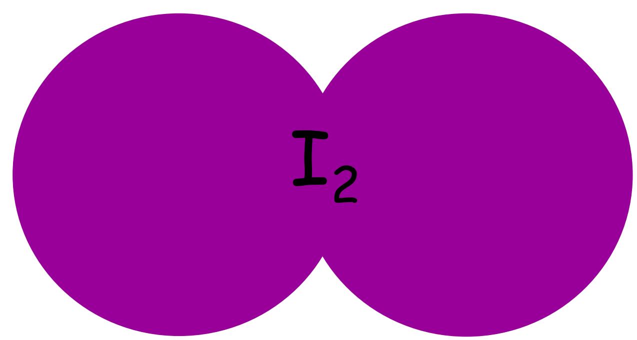


For C-C-C bond angle a change of 4° is typical. This moves a carbon atom about 10 pm.

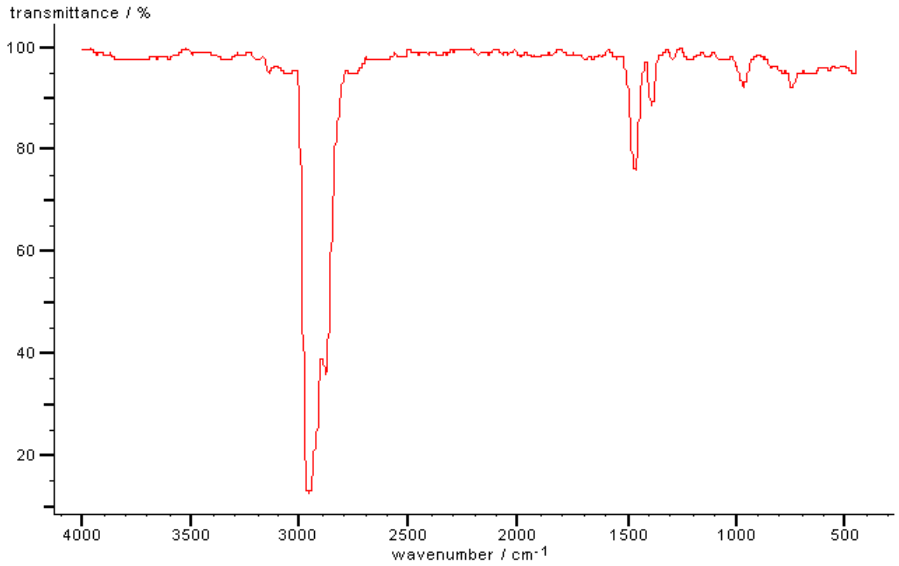
INFRARED SPECTROSCOPY

- Infrared (IR) electromagnetic radiation causes vibrations in molecules (wavelengths of 2500-15,000 nm or 2.5 – 15 mm)
- The greater the mass - the lower the wavenumber

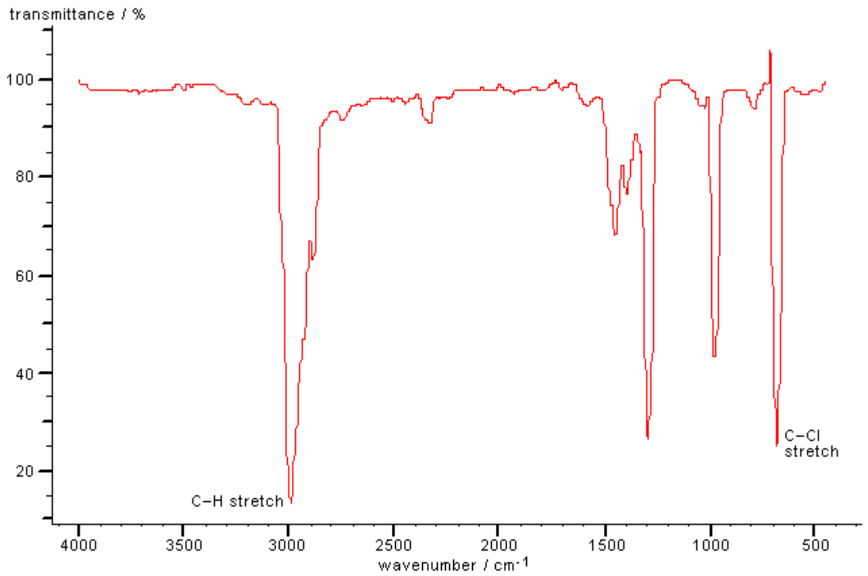




INFRARED SPECTROSCOPY



Ethane



Chloroethane

INFORMATION OBTAINED FROM IR SPECTRA

- IR is most useful in providing information about the presence or absence of specific **functional groups**.

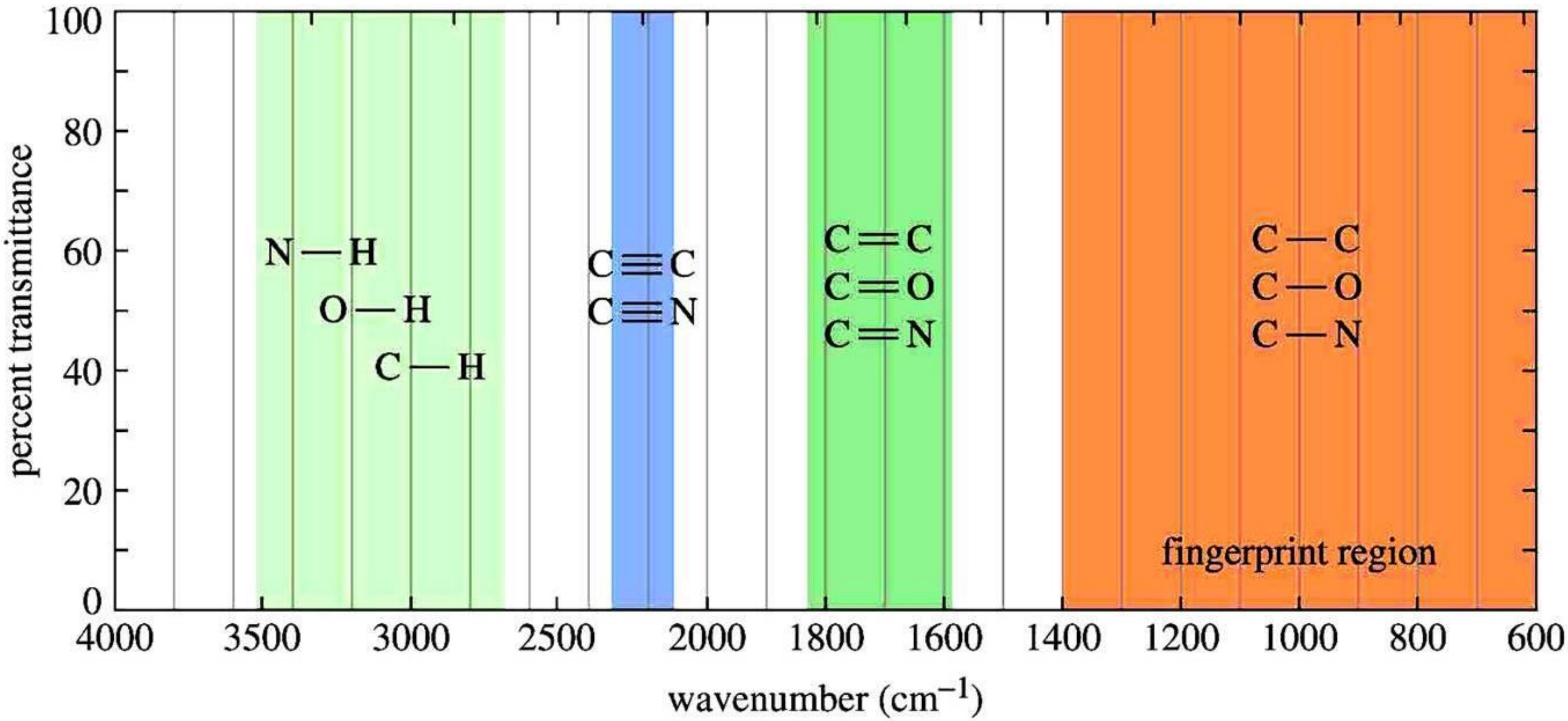
- IR can provide a **molecular fingerprint** that can be used when comparing samples. If two pure samples display the same IR spectrum it can be argued that they are the same compound.

- IR **does not** provide detailed information or proof of molecular formula or structure. It provides information on molecular fragments, specifically functional groups.

- Therefore it is very limited in scope, and must be used in conjunction with other techniques to provide a more complete picture of the molecular structure.

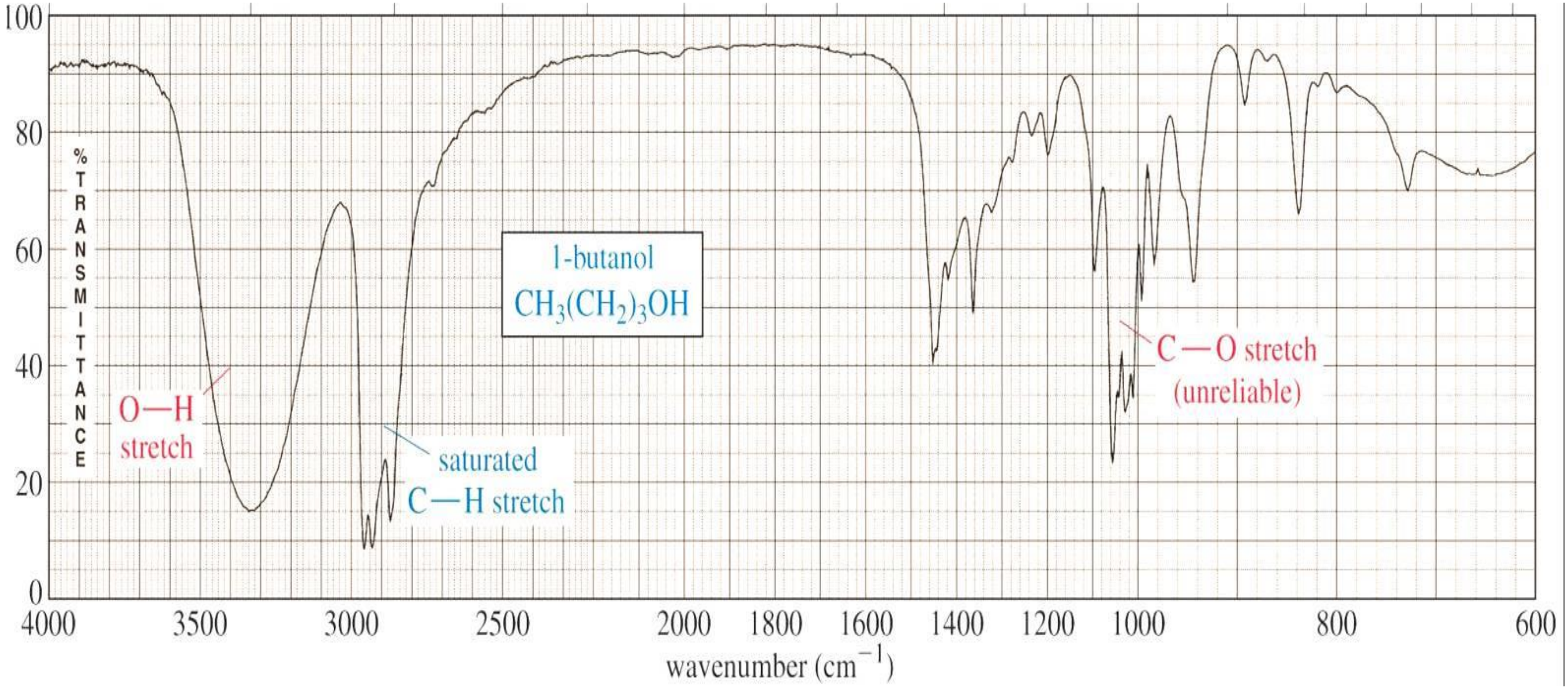
IR ABSORPTION RANGE

- The typical IR absorption range for covalent bonds is **600 - 4000 cm^{-1}**. The graph shows the regions of the spectrum where the following types of bonds normally absorb. For example a sharp band around 2200-2400 cm^{-1} would indicate the possible presence of a C-N or a C-C triple bond.



THE FINGERPRINT REGION

- Although the entire IR spectrum can be used as a fingerprint for the purposes of comparing molecules, the **600 - 1400 cm^{-1}** range is called the **fingerprint region**. This is normally a complex area showing many bands, frequently overlapping each other. This complexity limits its use to that of a fingerprint, thus neglect the region left of 1400 cm^{-1}.



Focus your analysis on this region. This is where most stretching frequencies appear.

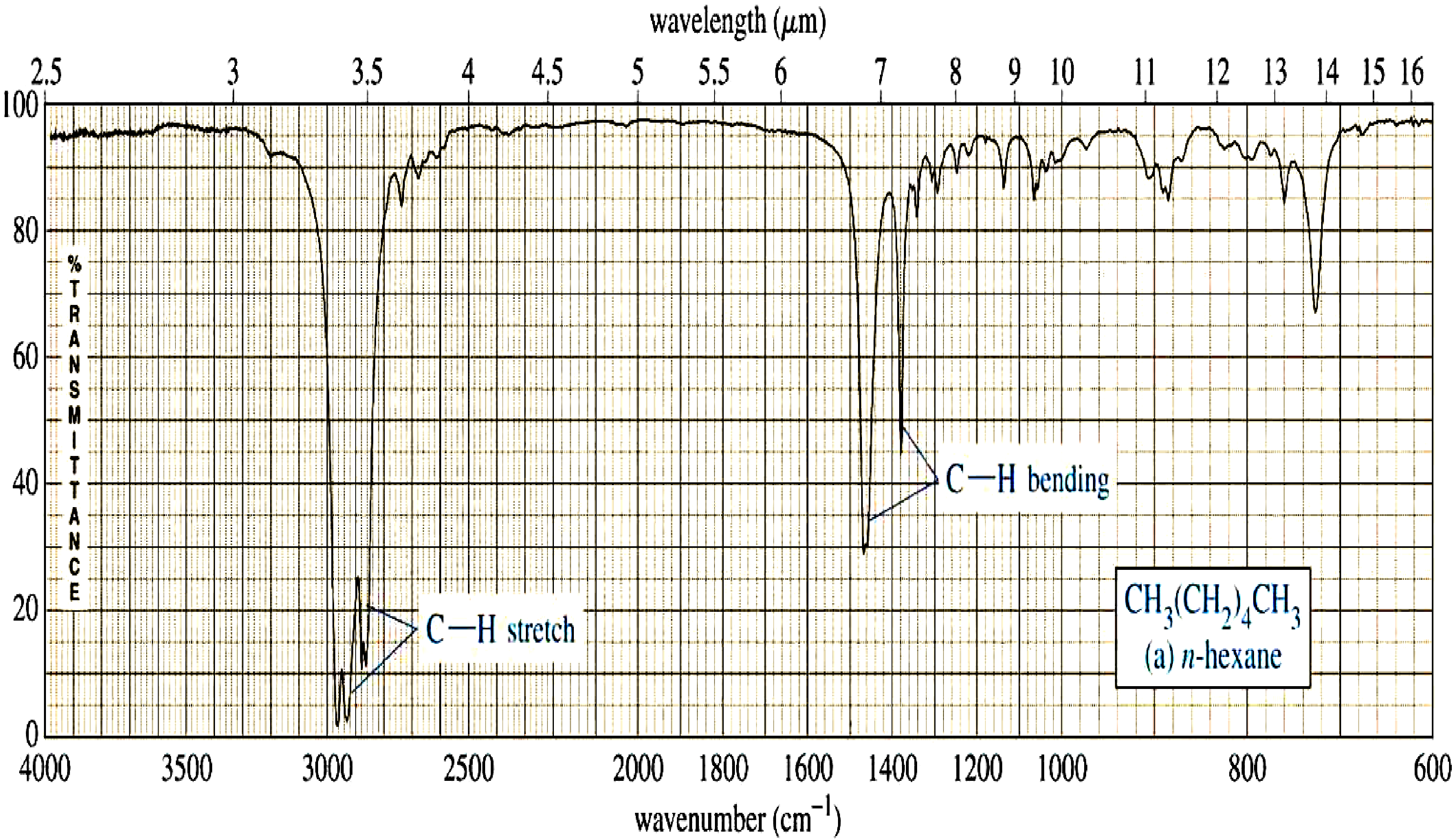
Fingerprint region: complex and difficult to interpret reliably.

Characteristic IR frequencies of common functional groups

Bond	Type of Compound	Frequency Range, cm^{-1}	Intensity
C—H	Alkanes	2850–2970	Strong
		1340–1470	Strong
C—H	Alkenes $\left(\gt C{=}C \lt^{H} \right)$	3010–3095	Medium
		675–995	Strong
C—H	Alkynes (—C≡C—H)	3300	Strong
C—H	Aromatic rings	3010–3100	Medium
		690–900	Strong
O—H	Monomeric alcohols, phenols	3590–3650	Variable
	Hydrogen-bonded alcohols, phenols	3200–3600	Variable, sometimes broad
	Monomeric carboxylic acids	3500–3650	Medium
	Hydrogen-bonded carboxylic acids	2500–2700	Broad
N—H	Amines, amides	3300–3500	Medium
C=C	Alkenes	1610–1680	Variable
C=C	Aromatic rings	1500–1600	Variable
C≡C	Alkynes	2100–2260	Variable
C—N	Amines, amides	1180–1360	Strong
C≡N	Nitriles	2210–2280	Strong
C—O	Alcohols, ethers, carboxylic acids, esters	1050–1300	Strong
C=O	Aldehydes, ketones, carboxylic acids, esters	1690–1760	Strong
NO_2	Nitro compounds	1500–1570	Strong
		1300–1370	Strong

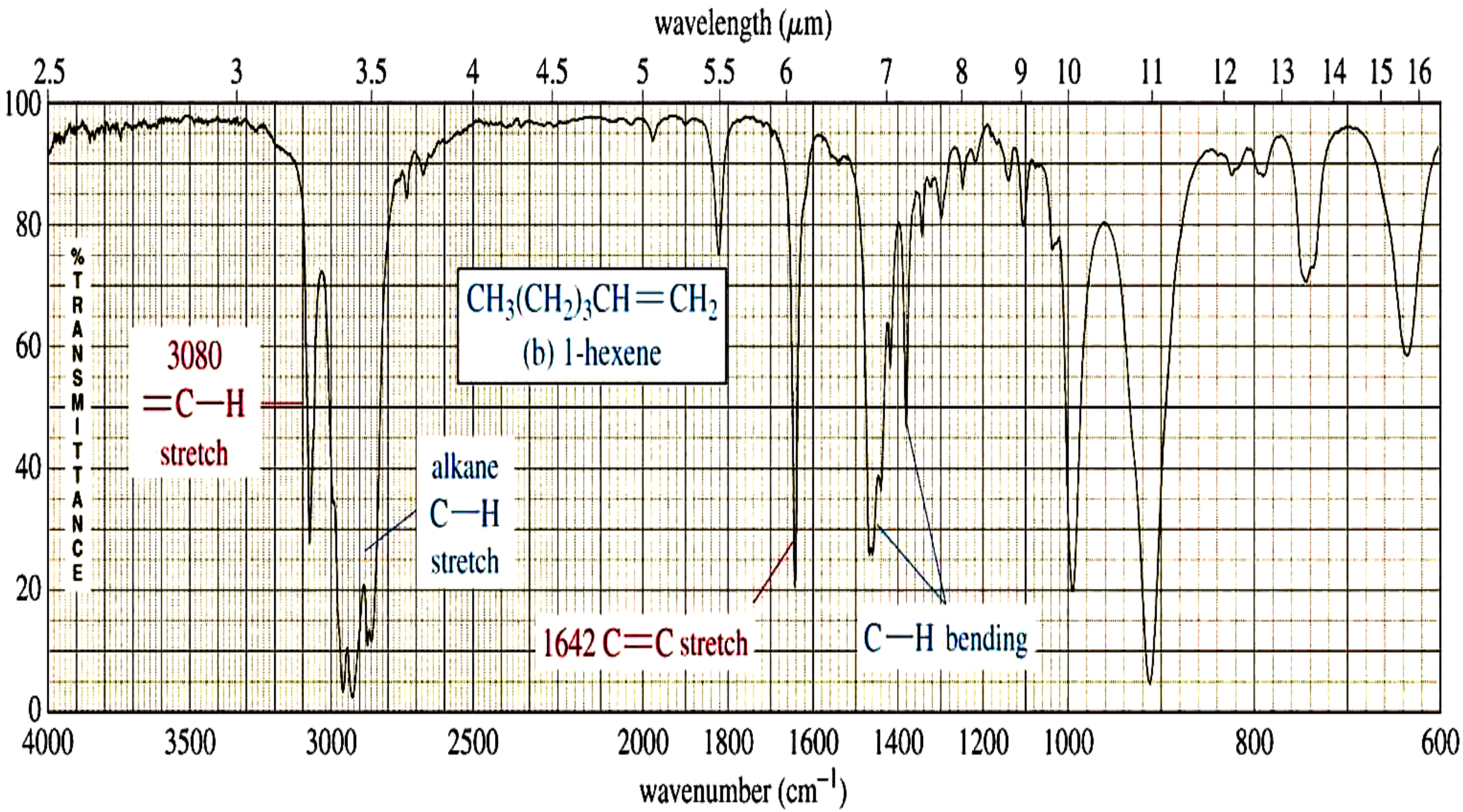
IR SPECTRUM OF ALKANES

- Alkanes have no functional groups. Their IR spectrum displays only C-C and C-H bond vibrations. Of these the most useful are the **C-H bands**, which appear around **3000 cm^{-1}**. Since most organic molecules have such bonds, most organic molecules will display those bands in their spectrum.



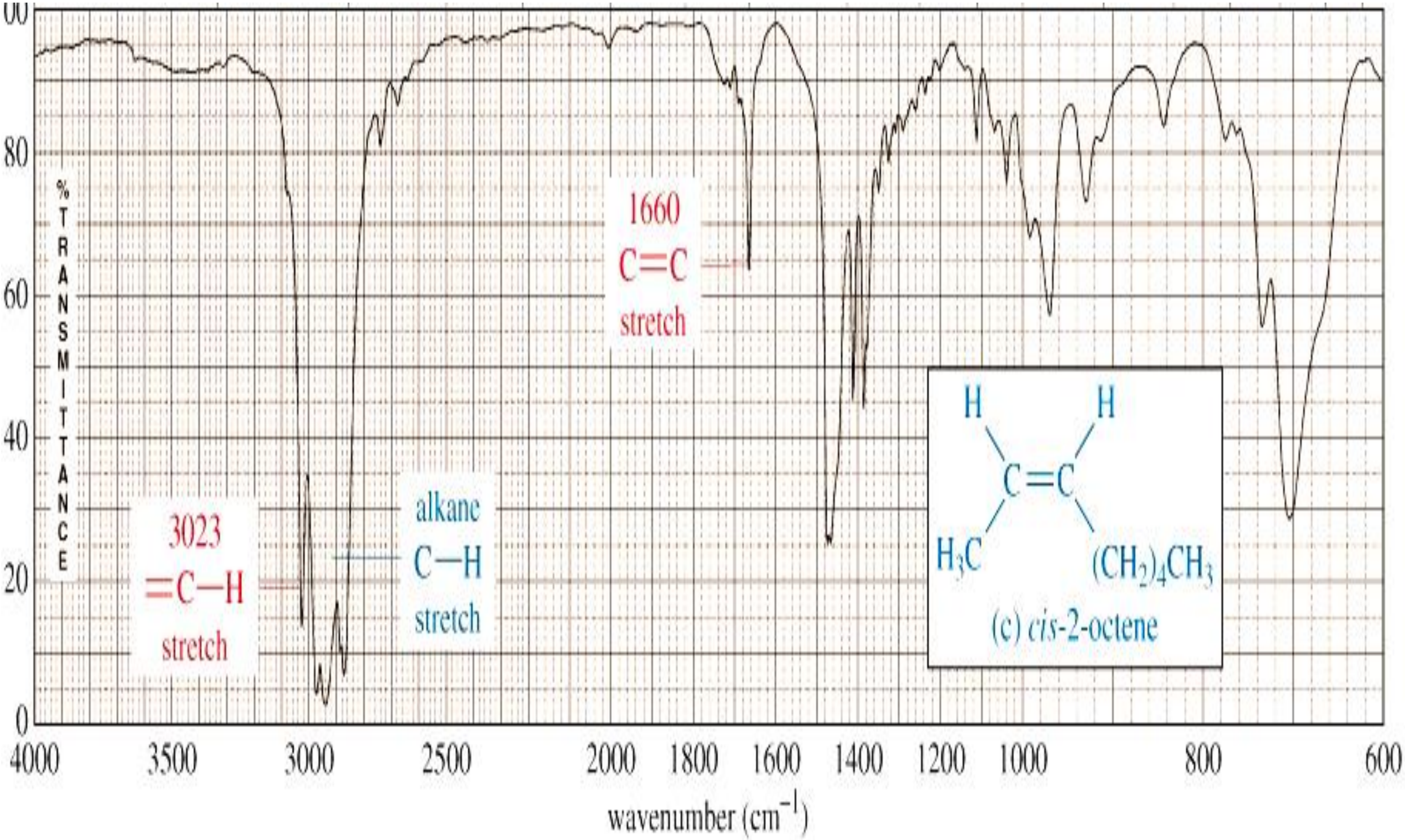
IR SPECTRUM OF ALKENES

- Besides the presence of C-H bonds, alkenes also show sharp, medium bands corresponding to the **C=C bond stretching vibration** at about **1600-1700 cm^{-1}**. Some alkenes might also show a band for the =C-H bond stretch, appearing around **3080 cm^{-1}** as shown below. However, this band could be obscured by the broader bands appearing around 3000 cm^{-1}



IR SPECTRUM OF ALKENES

- This spectrum shows that the band appearing around **3080 cm^{-1}** can be obscured by the broader bands appearing around 3000 cm^{-1}.



IR SPECTRUM OF ALKYNES

- The most prominent band in alkynes corresponds to the **carbon-carbon triple bond**.

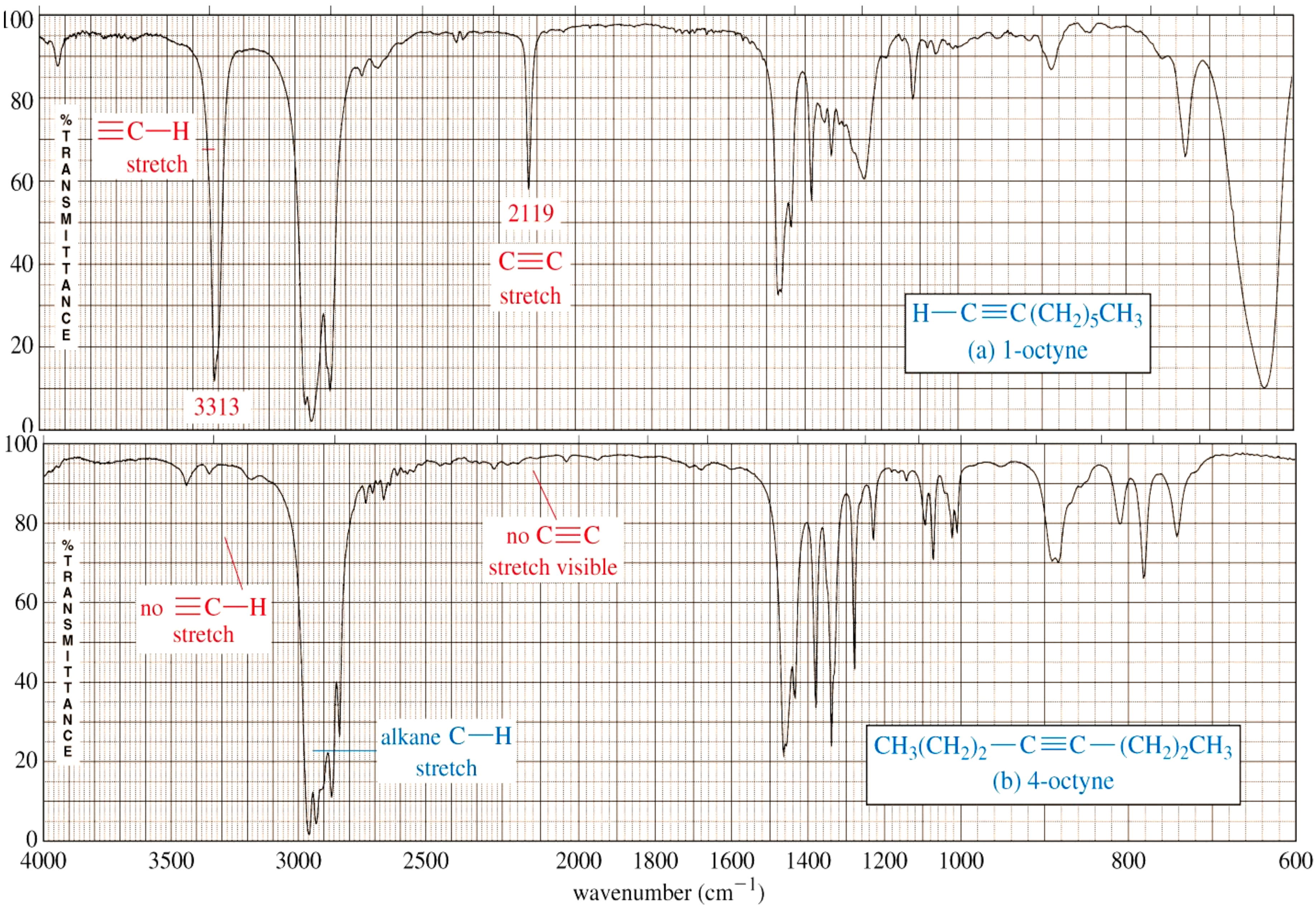
- It shows as a sharp, weak band at about **2100 cm^{-1}**. The reason it's weak is because the triple bond is not very polar.

- In some cases, such as in highly symmetrical alkynes, it may not show at all due to the low polarity of the triple bond associated with those alkynes.

- Terminal alkynes have C-H bonds involving the *sp* carbon (the carbon that forms part of the triple bond). Therefore they may also show a sharp, weak band at about **3300 cm^{-1}** corresponding to the C-H stretch.

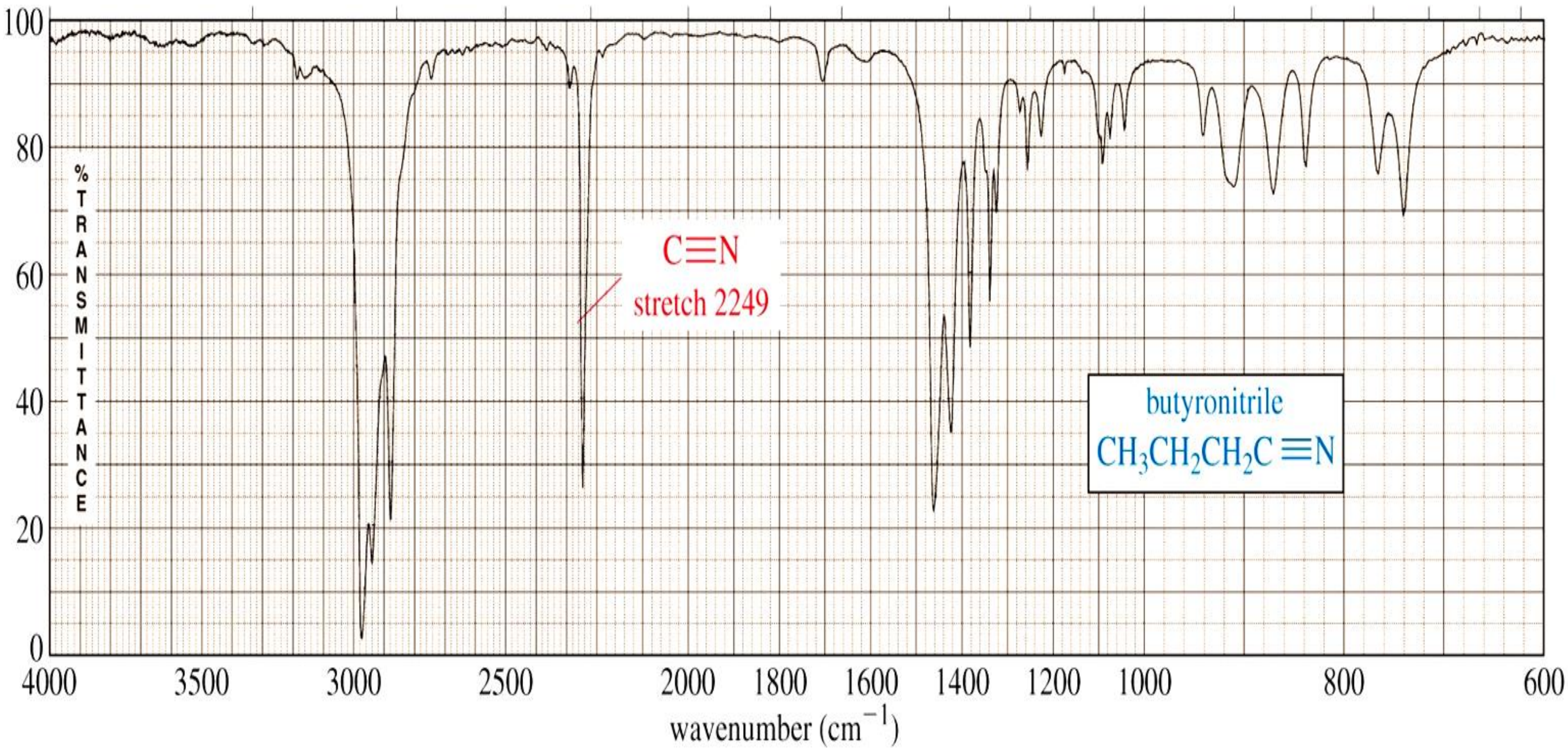
- Internal alkynes, that is those where the triple bond is in the middle of a carbon chain, do not have C-H bonds to the *sp* carbon and therefore lack the aforementioned band.

IR SPECTRUM OF ALKYNES



IR SPECTRUM OF A NITRILE

In a manner very similar to alkynes, nitriles show a prominent band around **2250 cm^{-1}** caused by the **CN triple bond**. This band has a sharp, pointed shape just like the alkyne C-C triple bond, but because the CN triple bond is more polar, this band is stronger than in alkynes.

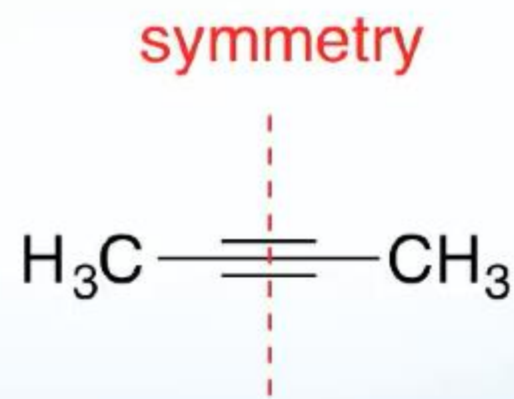


Signal Intensity in an IR Spectrum

For a bond to absorb in IR, there must be a change in the dipole moment during the vibration.

If a bond is non-polar due to symmetry, IR light will not be absorbed and the vibration is said to be IR inactive.

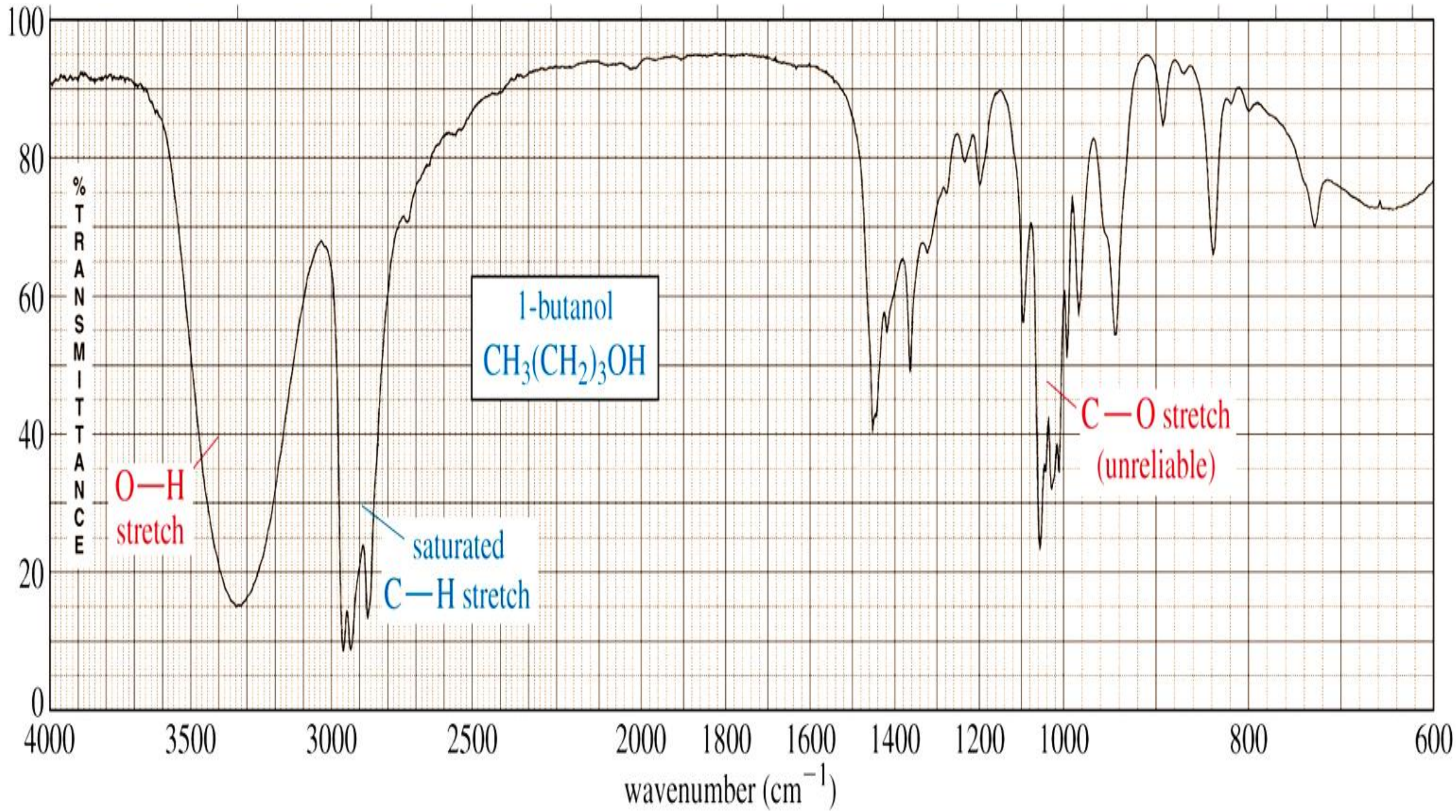
C≡C is IR active



C≡C is IR inactive

IR SPECTRUM OF AN ALCOHOL

- The most prominent band in alcohols is due to the **O-H bond**, and it appears as a strong, broad band covering the range of about **3000 - 3700 cm^{-1}**. The sheer size and broad shape of the band dominate the IR spectrum and make it hard to miss.



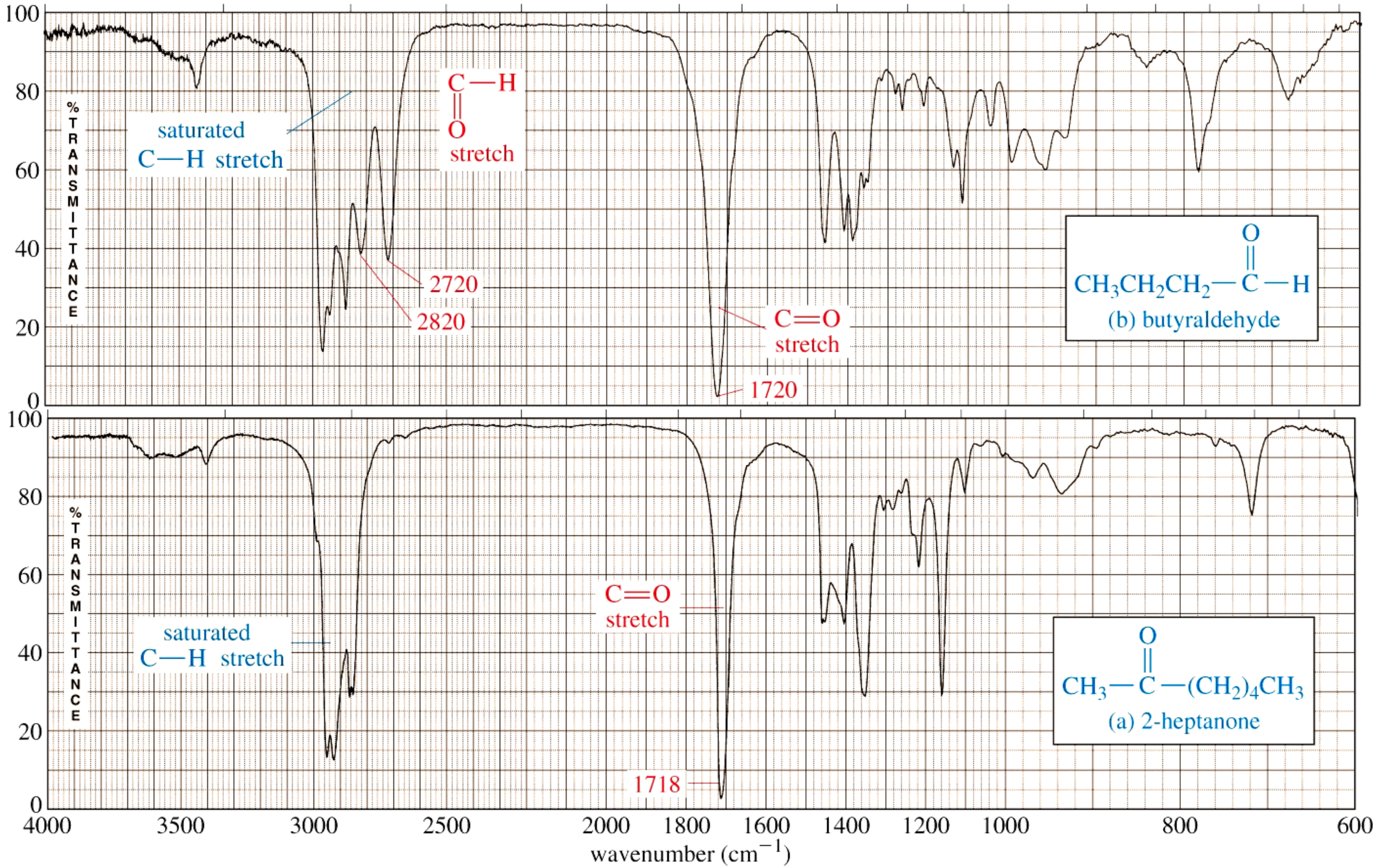
IR SPECTRUM OF ALDEHYDES AND KETONES

- Carbonyl compounds **contain C=O functional group**.

- Aldehydes and ketones show a strong, prominent, stake-shaped band around **1710 - 1720 cm^{-1}** (right in the middle of the spectrum). This band is due to the **highly polar C=O bond**. Because of its position, shape, and size, it is hard to miss.

- Because aldehydes also contain a C-H bond to the sp^2 carbon of the C=O bond, they also show a pair of medium strength bands positioned about **2700** and **2800 cm^{-1}**. These bands are missing in the spectrum of a ketone because the sp^2 carbon of the ketone lacks the C-H bond.

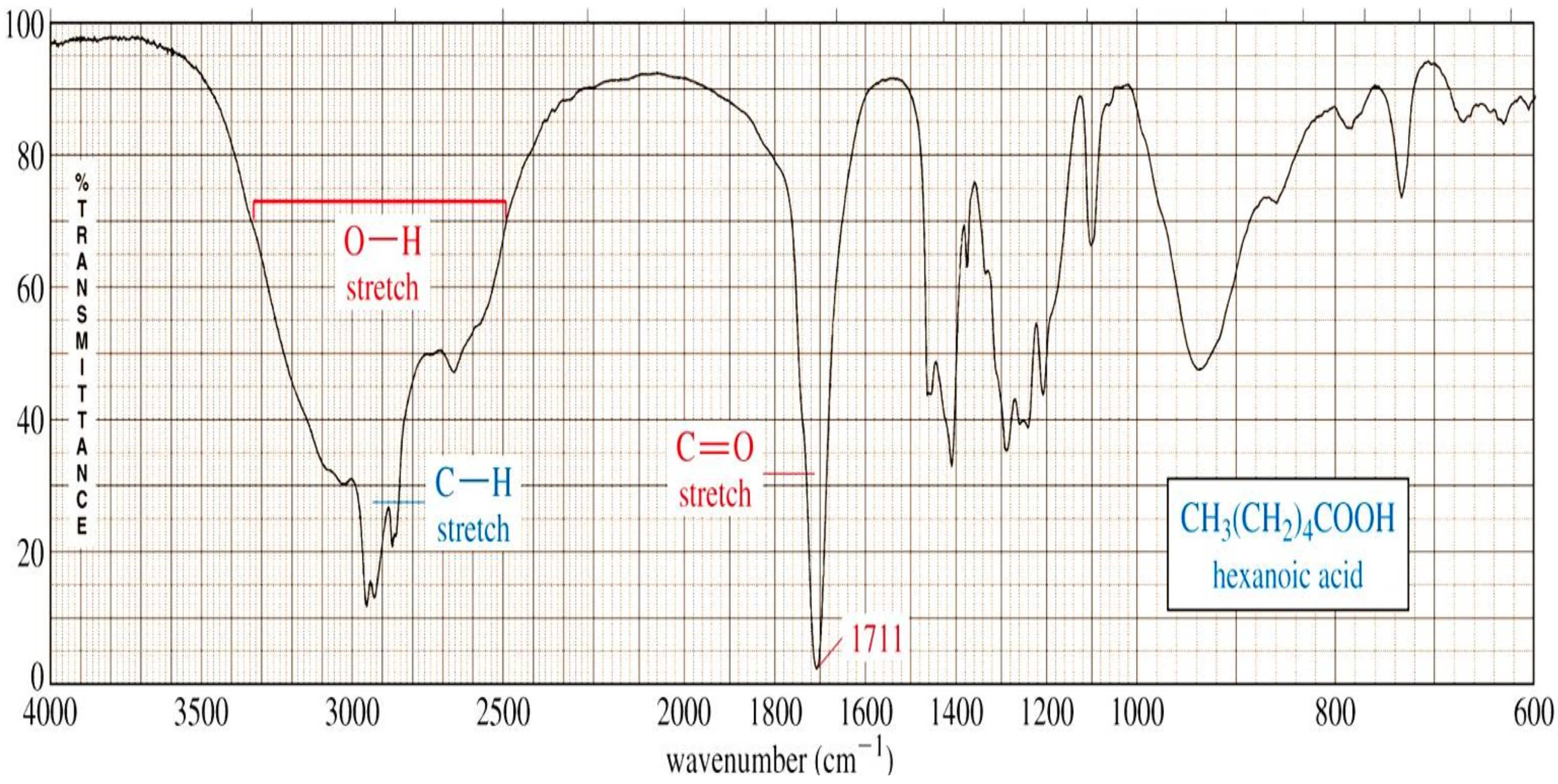
IR SPECTRUM OF ALDEHYDES AND KETONES



Graphics source: Wade, Jr., L.G. *Organic Chemistry*, 6th ed. Pearson Prentice Hall Inc., 2006

IR SPECTRUM OF A CARBOXYLIC ACID

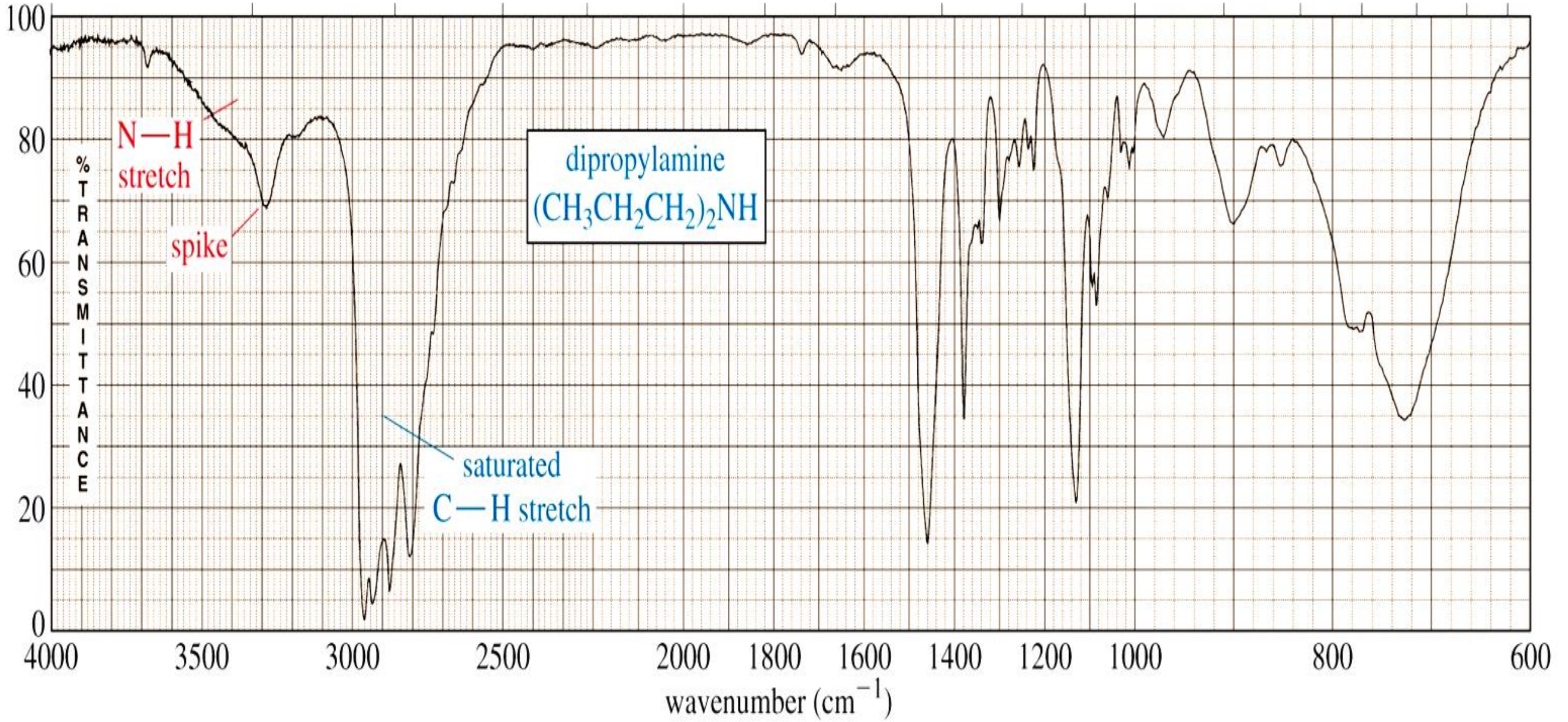
- A carboxylic acid functional group combines the features of alcohols and ketones because it has both the **O-H bond** and the **C=O bond**. Therefore carboxylic acids show a very strong and broad band covering a wide range between **2800** and **3500 cm^{-1}** for the O-H stretch. At the same time they also show the stake-shaped band in the middle of the spectrum around **1710 cm^{-1}** corresponding to the C=O stretch.



IR SPECTRA OF AMINES

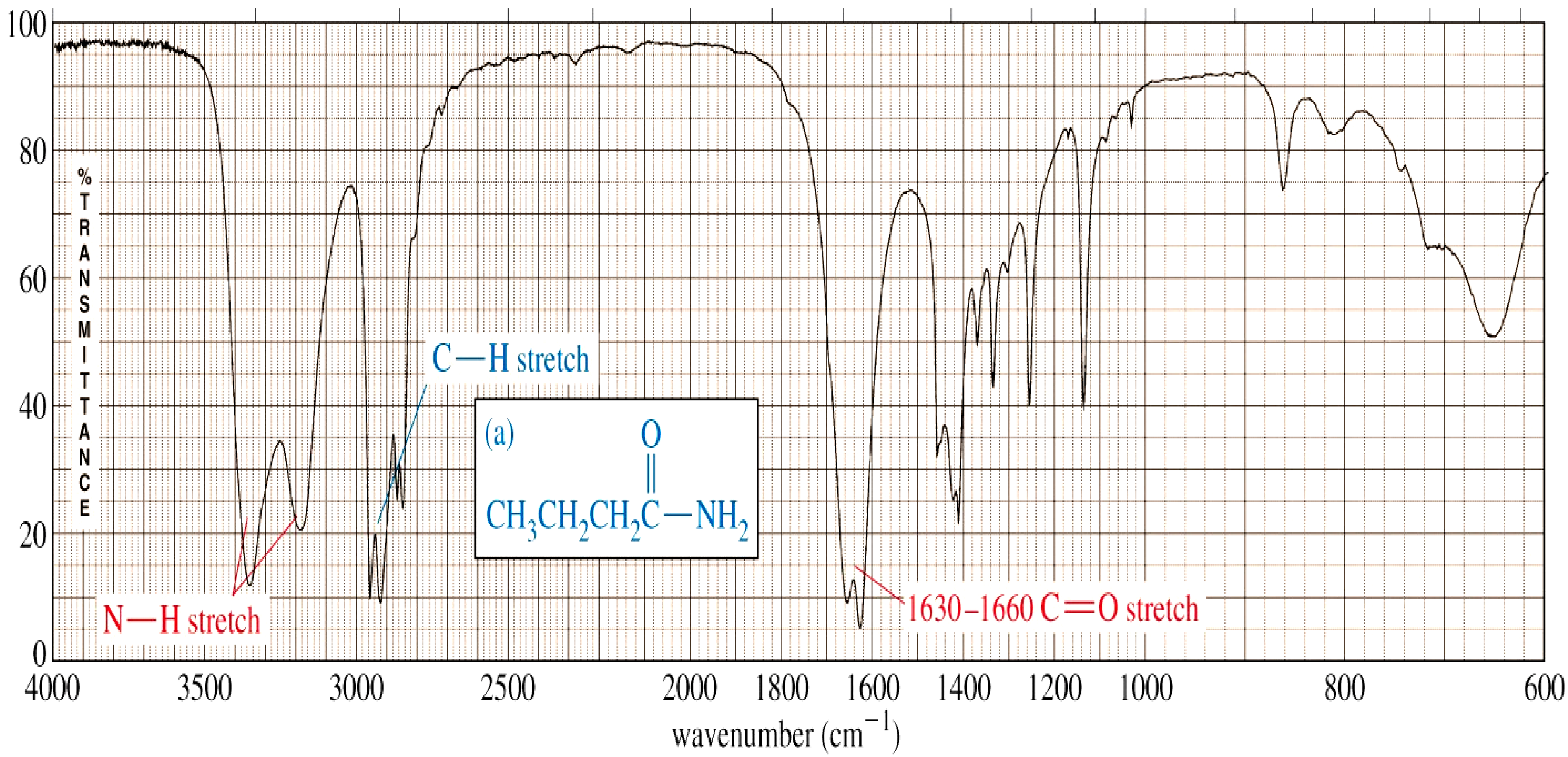
The most characteristic band in amines is due to the **N-H bond stretch**, and it appears as a weak to medium, somewhat broad band (but not as broad as the O-H band of alcohols). This band is positioned at the left end of the spectrum, in the range of about **3200 - 3600 cm^{-1}**.

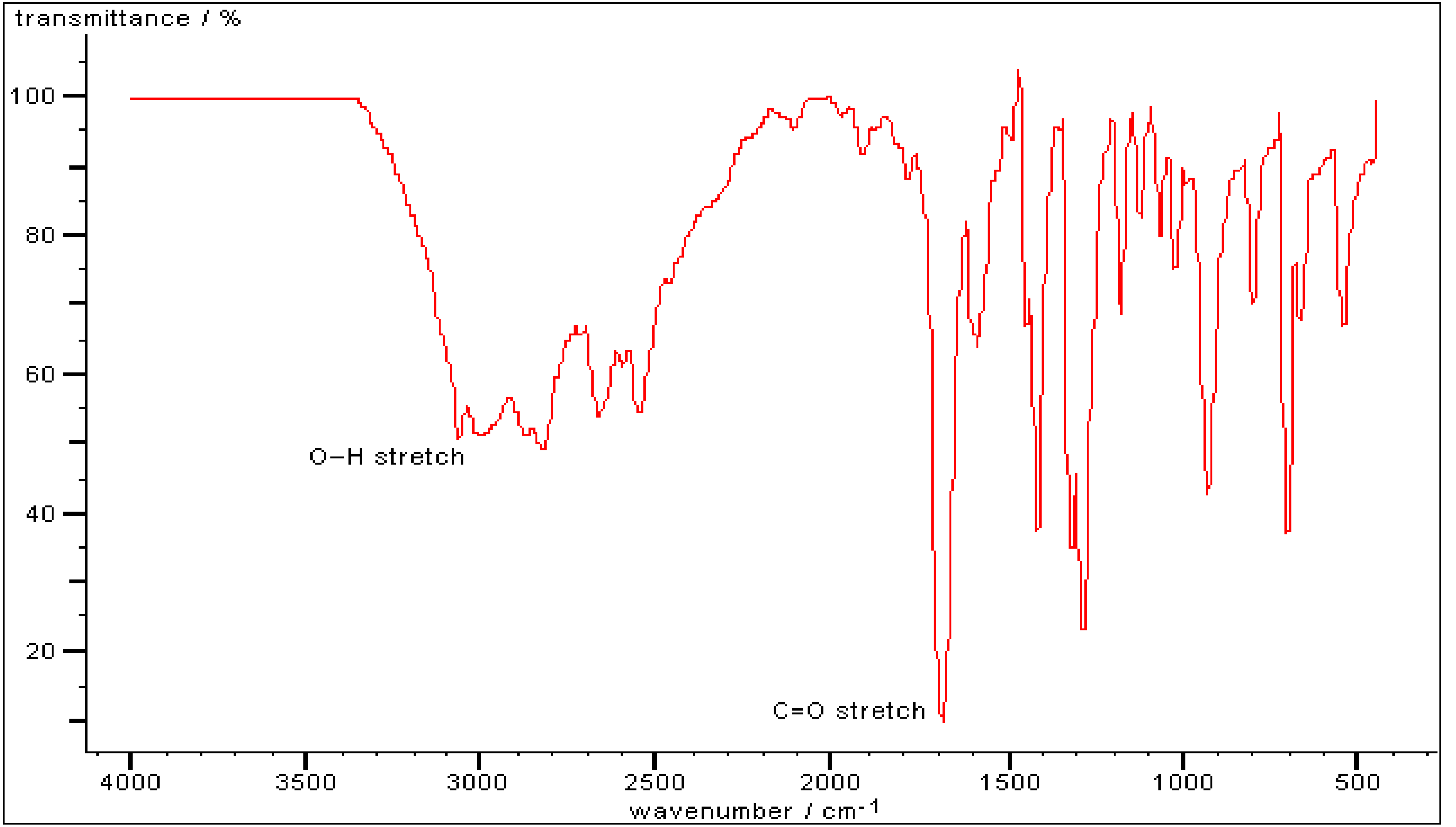
Primary amines have two N-H bonds, therefore they typically show two spikes that make this band resemble a molar tooth. Secondary amines have only one N-H bond, which makes them show only one spike, resembling a canine tooth. Finally, tertiary amines have no N-H bonds, and therefore this band is absent from the IR spectrum altogether. The spectrum below shows a secondary amine.



IR SPECTRUM OF AMIDES

The amide functional group combines the features of amines and ketones because it has both the **N-H bond** and the **C=O bond**. Therefore amides show a very strong, somewhat broad band at the left end of the spectrum, in the range between **3100** and **3500 cm^{-1}** for the N-H stretch. At the same time they also show the stake-shaped band in the middle of the spectrum around **1710 cm^{-1}** for the C=O stretch. As with amines, primary amides show two spikes, whereas secondary amides show only one spike.





Benzoic acid

INFRARED SPECTROSCOPY

POSITION	**REDUCED MASS** **BOND STRENGTH (STIFFNESS)**	**LIGHT ATOMS HIGH FREQUENCY** **STRONG BONDS HIGH FREQUENCY**
STRENGTH	**CHANGE IN ‘POLARITY’**	**STRONGLY POLAR BONDS GIVE INTENSE BANDS**
WIDTH	**HYDROGEN BONDING**	**STRONG HYDROGEN BONDING GIVES BROAD BANDS**

INFRARED SPECTROSCOPY

- In general

Bond →	C-H	C-D	C-O	C-Cl
ν/cm^{-1} →	3000	2200	1100	700
Bond →	C≡O	C=O	C-O	
ν/cm^{-1} →	2143	1715	1100	

INFRARED SPECTROSCOPY

4000-3000 cm^{-1}	3000-2000 cm^{-1}	2000-1500 cm^{-1}	1500-1000 cm^{-1}
O-H N-H C-H	C≡C C≡N	C=C C=O	C-O C-F C-Cl deformations

Increasing energy

Increasing frequency

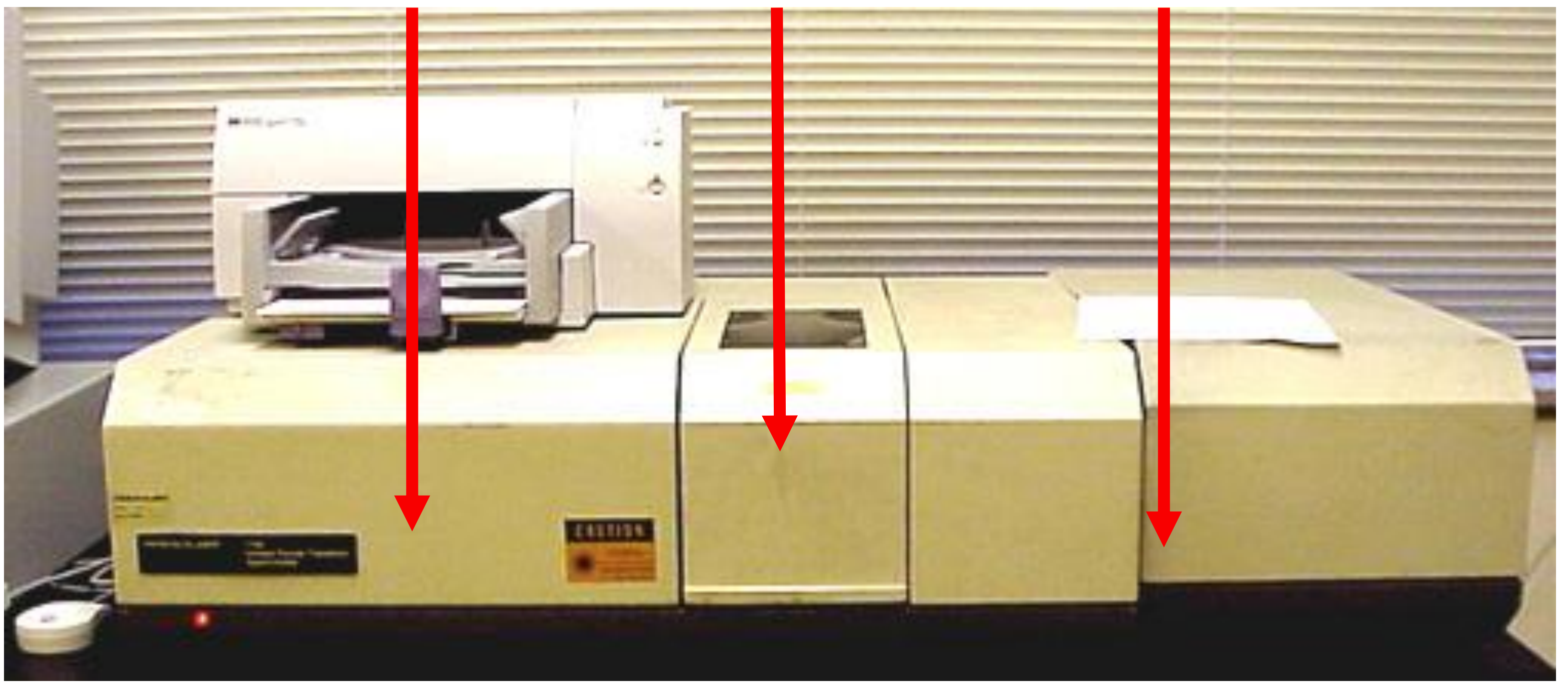
INTERPRETATION OF INFRARED SPECTRA

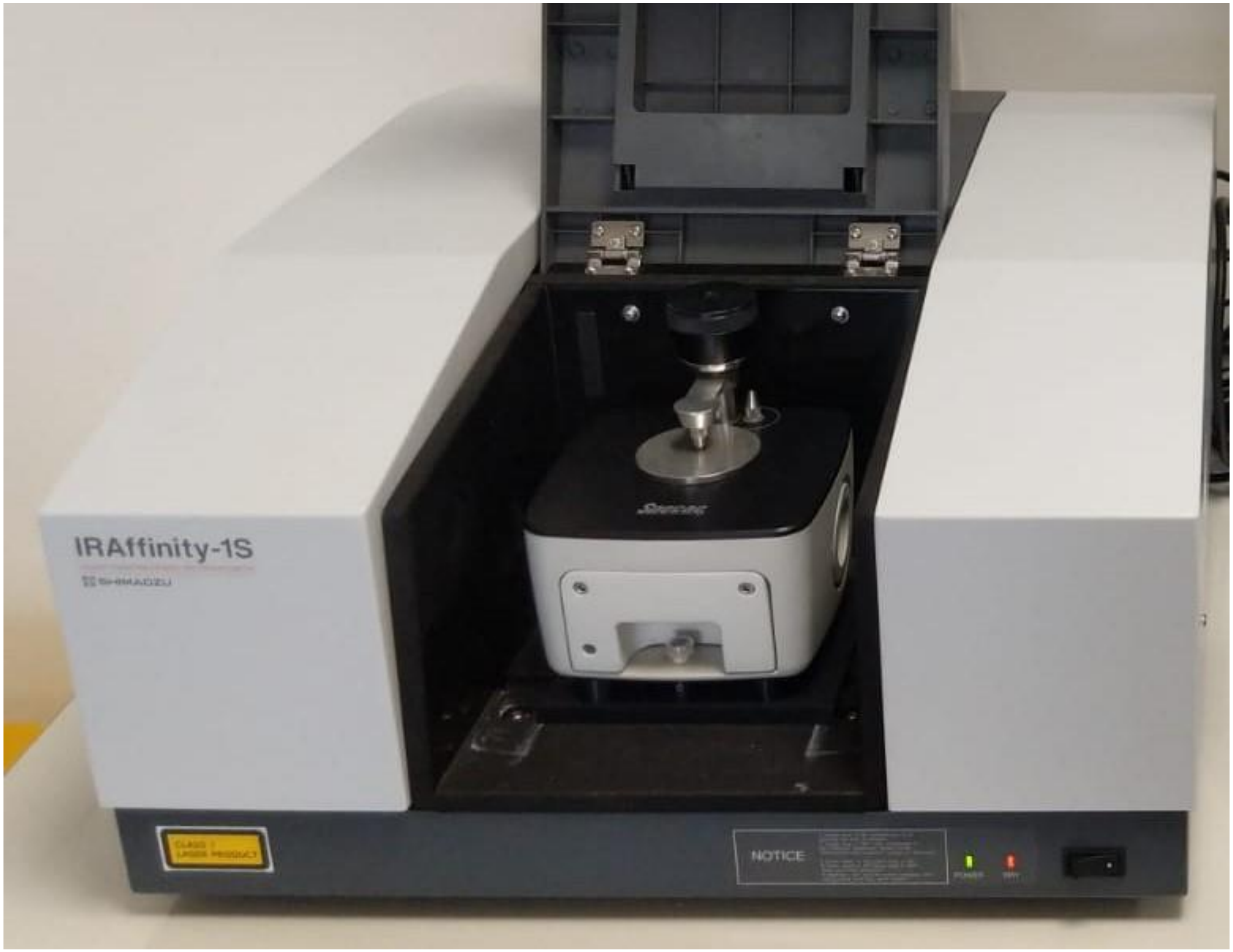
- An element of judgement is required in interpreting IR spectra but you should find that it becomes relatively straightforward with practice.

- It is often possible to assign the peaks in the 1600-3600 cm^{-1} region by consulting tables or databases of IR spectra. When making an assignment, give both the type of bond and the type of vibration, *e.g.* O-H stretch or C-H bending vibration.
- The most useful regions are as follows:
  - 1680-1750 cm^{-1}:C=O stretches feature very strongly in IR spectra and the type of carbonyl group can be determined from the exact position of the peak.
  - 2700-3100 cm^{-1}: different types of C-H stretching vibrations.
  - 3200-3700 cm^{-1}: various types of O-H and N-H stretching vibrations.

- Too many bonds absorb in the region of 600-1600 cm^{-1} to allow confident assignment of individual bands. However, this region is useful as a fingerprint of a molecule, *i.e.* if the spectrum is almost identical to an authentic reference spectrum then the structure can be assigned with some confidence.

Infrared Instrumentation

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NOTICE

- **Dispersive instruments:** with a monochromator to be used in the mid-IR region for spectral scanning and quantitative analysis.

- **Fourier transform IR (FTIR) systems:** widely applied and quite popular in the far-IR and mid-IR spectrometry.

- **Nondispersive instruments:** use filters for wavelength selection or an infrared-absorbing gas in the detection system for the analysis of gas at specific wavelength.

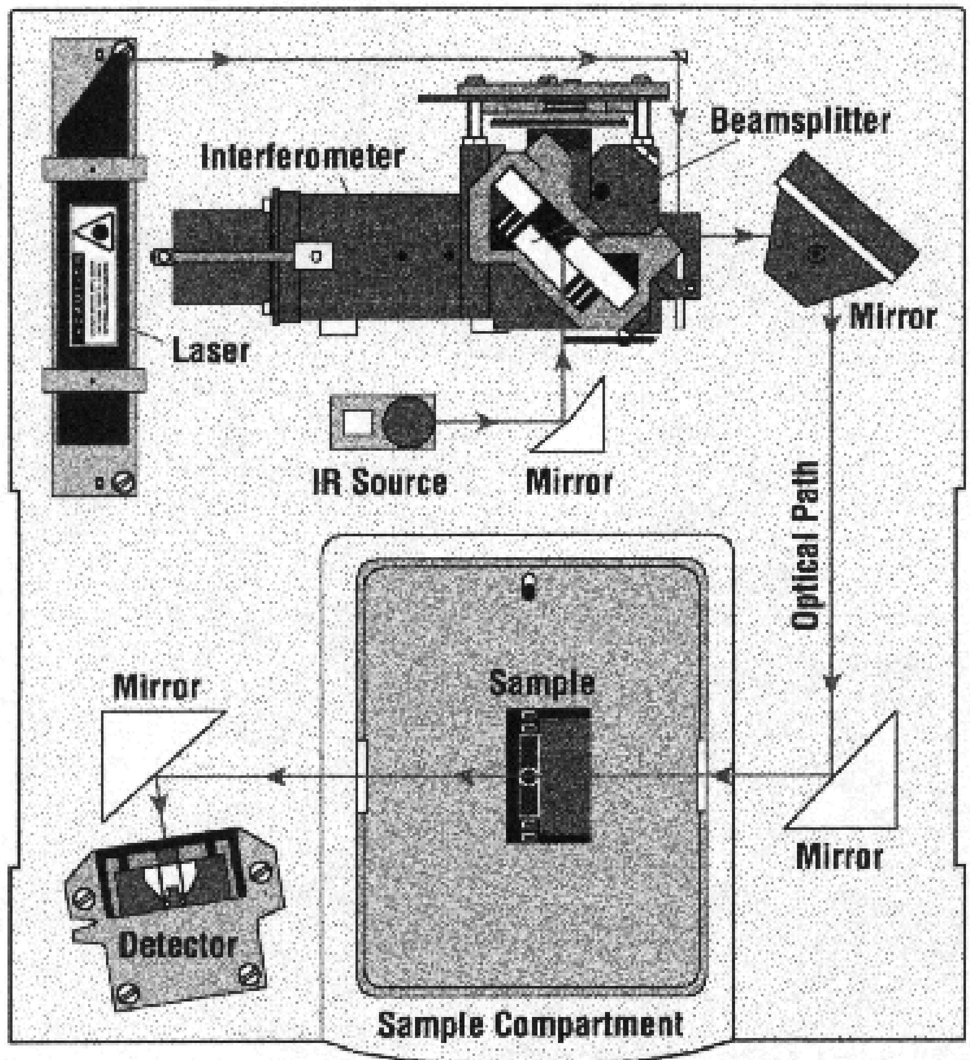
Infrared Instrumentation

- All modern instruments are Fourier Transform instruments.

- In all transmission experiments radiation from a source is directed through the sample to a detector.

- The measurement of the type and amount of light transmitted by the sample gives information about the structure of the molecules comprising the sample.

Infrared Instrumentation



Infrared Experimental

- To obtain an IR spectrum, the sample must be placed in a “container” or cell that is transparent in the IR region of the spectrum.

- Sodium chloride or salt plates are a common means of placing the sample in the light beam of the instrument.

InfraredExperimental

IR transparent Salt Plates

Infrared Experimental

- These plates are made of salt and must be stored in a **<u>water free</u>** environment such as the dessicator shown here.

Infrared Experimental

- The plates must also be handled with gloves to avoid contact of the plate with moisture from one’s hands.

Infrared Experimental

- To run an IR spectrum of a liquid sample, a drop or two of the liquid sample is applied to a salt plate.

- A second salt plate is placed on top of the first one such that the liquid forms a thin film “sandwiched” between the two plates.

Infrared Experimental

- The cell holder is then placed in the beam of the instrument.
- The sample is then scanned by the instrument utilizing predestinated parameters.
- A satisfactory spectrum has well defined peaks-but not so intense as to cause flattening on the bottom of the peaks.

Infrared Experimental

- The salt plates are cleaned by rinsing into a waste container with a suitable organic solvent-commonly cyclohexane - NEVER WATER!

- Cloudy plates must be polished to return them to a transparent condition.

- To polish cloudy windows, rotate salt plate on polishing cloth.